To utilize this implementation, you need to modify the import statement in ch07/rag-web-app/src/app/(chat)/actions.jsx to

```
import { LogProbsRag as RAG } from '@/lib/LogPropsRag';
```

This adjustment enables the use of the new `performEnhancedRAG` method, which incorporates log probability grounding for improved response accuracy and reliability. The details of the `LogPropsRag` file are not shown here, but essentially it uses an AI-powered confidence evaluation process. This evaluation examines whether the retrieved context contains sufficient information to comprehensively answer the user's question. When the confidence level is high, the method proceeds with generating a response using the retrieved documents. However, if the confidence falls below a specified threshold, the method activates a fallback strategy. In that case, it will ask the user to rephrase or provide more context about their query.

> **WARNING** Grounding support can significantly increase the number of API requests made during interactions with AI models. When grounding is enabled, each query may trigger additional requests to retrieve relevant external data, such as information from Google Search or requesting log probabilities. This can lead to higher usage costs and potential rate limiting if the API has a daily query limit. Additionally, grounding may create challenges with large context windows, as incorporating extensive external data could exceed the model's input size limitations.

With those changes in place, we can now rerun the application and try to ask the RAG system a question that does not have a valid answer because it's not part of the knowledge base. Let's examine how the AI's responses differ when it encounters questions about topics outside its knowledge base, both with and without proper grounding context. Figure 7.11 shows the importance of context in preventing hallucination and ensuring accurate responses.

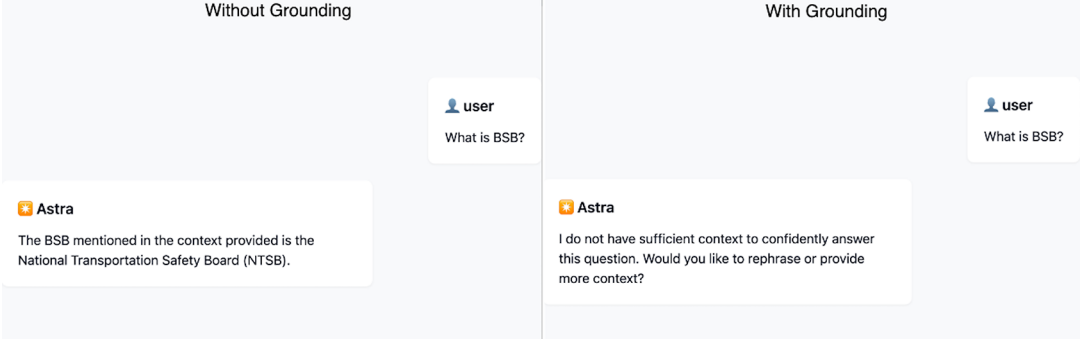


Figure 7.11 Comparison of AI responses with and without grounding context. The left panel shows the AI's response without grounding, while the right panel shows the AI's response when provided with additional contextual information to ground its understanding.